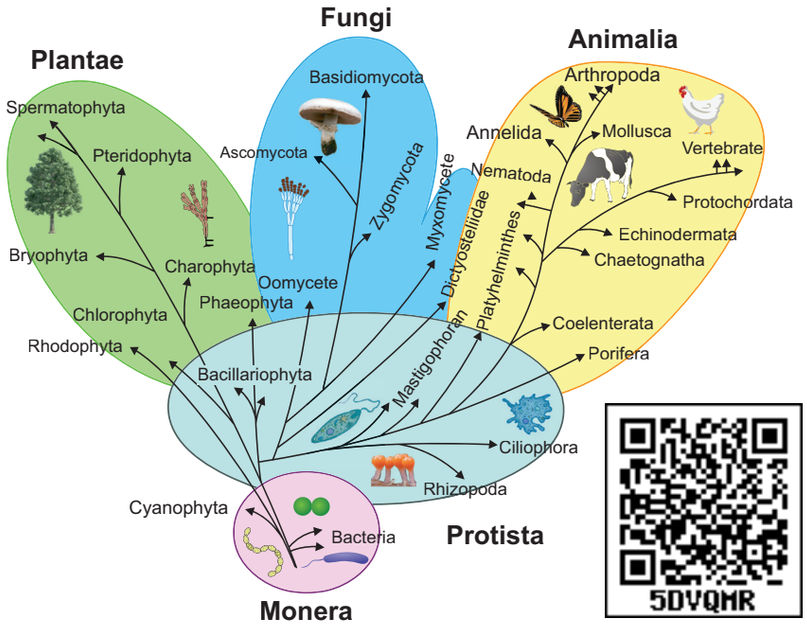



**Figure 1.7:** Five Kingdom Classification

**Protista, Fungi, Plantae and Animalia** (Figure 1.7). The criteria adopted for the classification include cell structure, thallus organization, mode of nutrition, reproduction and phylogenetic relationship**.** A comparative account of the salient features of each Kingdom is given in Table 1.5

**Merits**

- The classification is based on the complexity of cell structure and organization of thallus.
- It is based on the mode of nutrition
- Separation of fungi from plants
- It shows the phylogeny of the organisms

**Demerits**

- The Kingdom Monera and protista accommodate both autotrophic and heterotrophic organisms, cell wall lacking and cell wall bearing organisms thus making these two groups more heterogeneous.
- Viruses were not included in the system.

Carl Woese and co-workers in the year 1990 introduced three domains of life *viz.,* **Bacteria**, **Archaea** and **Eukarya** based on the difference in rRNA nucleotide sequence, lipid structure of the cell membrane. A revised six Kingdom classification for living world was proposed by Thomas Cavalier-Smith in the year 1998 and the Kingdom **Monera** is divided in to **Archaebacteria** and **Eubacteria.** Recently Ruggierio *et al.*, 2015 published a seven Kingdom classification which is a practical extension of Thomas Cavalier's six Kingdom scheme. According to this classification there are two Super Kingdoms. (**Prokaryota** and **Eukaryota**) **Prokaryota** includes two Kingdoms namely **Archaebacteria** and **Eubacteria.** Eukaryota includes the **Protozoa, Chromista, Fungi, Plantae** and **Animalia.** A new Kingdom, the **Chromista** was erected and it included all algae whose chloroplasts contain chlorophyll a and c, as well as various colourless forms that are closely related to them. Diatoms, Brown algae, Cryptomonads and Oomycetes were placed under this Kingdom.

### Activity 1.2

Visit to a pond and record the names of the biotic components of it with the help of your teacher. Tabulate the data and segregate them according to Five Kingdom Classification.




Red tide is caused by toxic bloom of Dinoflagellates like *Gymnodinium breve* and *Gonyaulax tamarensis*.

A major red tide incident in west coast of Florida in the year 1982 killed Hundreds and thousands of fishes.



## 1.4 Bacteria

### *Bacteria Friends or Foes?*

Have you noticed the preparation of curd in our home? A little drop of curd turns the milk into curd after some time. What is responsible for this change? Why it Sours? The change is brought by *Lactobacillus lactis*, a bacterium present in the curd. The sourness is due to the formation of Lactic



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